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First report of coupled Early Permian paleomagnetic and geochronologic data from the Dunhuang block (NW China), and implications for the tectonic evolution of the Paleo-Asian ocean

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ABSTRACT

In order to better understand the tectonic relationship between the Dunhuang block (DHB) and its adjacent blocks, and to constrain the timing of the closure of the Paleo-Asian ocean, a combined geochronologic and paleomagnetic investigation has been undertaken on Early Permian tuff, basalt flows and sandstones from the Shuangbaotang Formation in the northwestern part of the DHB. U-Pb zircon dating indicates that the age of the strata is between 280.6 ± 2.9 Ma and 291.4 ± 2.6 Ma. Thermal demagnetization of a three-component isothermal remanent magnetization (IRM), and Curie point experiment suggest that magnetite dominates in the rock samples analyzed. In addition, there is a minor amount of hematite in some sandstones. Stepwise thermal demagnetization successfully isolated stable characteristic remanent magnetization (ChRM) from 11 tuff layers, two lava flows and nine sandstone beds. Two components were isolated from all samples: a lower temperature component (LTC) and a higher temperature component (HTC). The LTC is near the direction of the present-day geomagnetic field and produced a negative fold test, indicating it is a viscous remanent magnetization in the present-day geomagnetic field. Most of the HTC are reverse polarity (Normal = 4, reverse = 158), which is in accordance with the Kiaman Reversed Superchron that spans the Late Carboniferous-Permian interval. In addition, the HTC of all studied sites passed the fold and reverse tests, suggesting that they likely represent primary remanent magnetization. The tilt-corrected mean direction from all sites (tuff, basalts and sandstones) is D_s = 1.7°, I_s = 43.1°, k_s = 403, α_{95} = 1.5°, N = 22. The mean paleopole of the site-mean direction-corresponded VGPs lies at 74.5°N, 268.5°E with A_{95} = 1.6°. Considering the consistent inclination values recorded among the studied tuff, basalts and sandstones, and the low degrees of anisotropy within all samples, we suggest that there is no significant inclination shallowing caused by depositional compaction in the sedimentary layers of the studied section. Taking into account the results from this study as well as previous Late Paleozoic paleomagentic studies from adjacent tectonic blocks, we conclude that the DHB formed part of the amalgamated Dunhuang-North China-Alxa-Qaidam mega-block during the Early Permian, but was separated from the Tarim block by a small ocean (here named the Qiemo-Xingxingxia paleo-ocean) at this time. A comparison of the Early Permian paleolatitudes of these and other adjacent blocks suggest that the Paleo-Asian ocean (sensu *lato*) was still open at this time. Combined with other geological evidence, a paleogeographic reconstruction of the Paleo-Asian ocean has been reconstructed for the Early Permian.

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1. Introduction

The Central Asian orogenic belt (CAOB), extending from the northern margins of the Tarim block and North China block (NCB) to the southern margin of the Siberian block (Fig. 1a), is generally understood to have resulted from the subduction and closure of the Paleo-Asian ocean, through the long-term tectonic evolution and final amalgamation of the Siberian, Tarim, NCB, Mongolia-Tuva blocks and several

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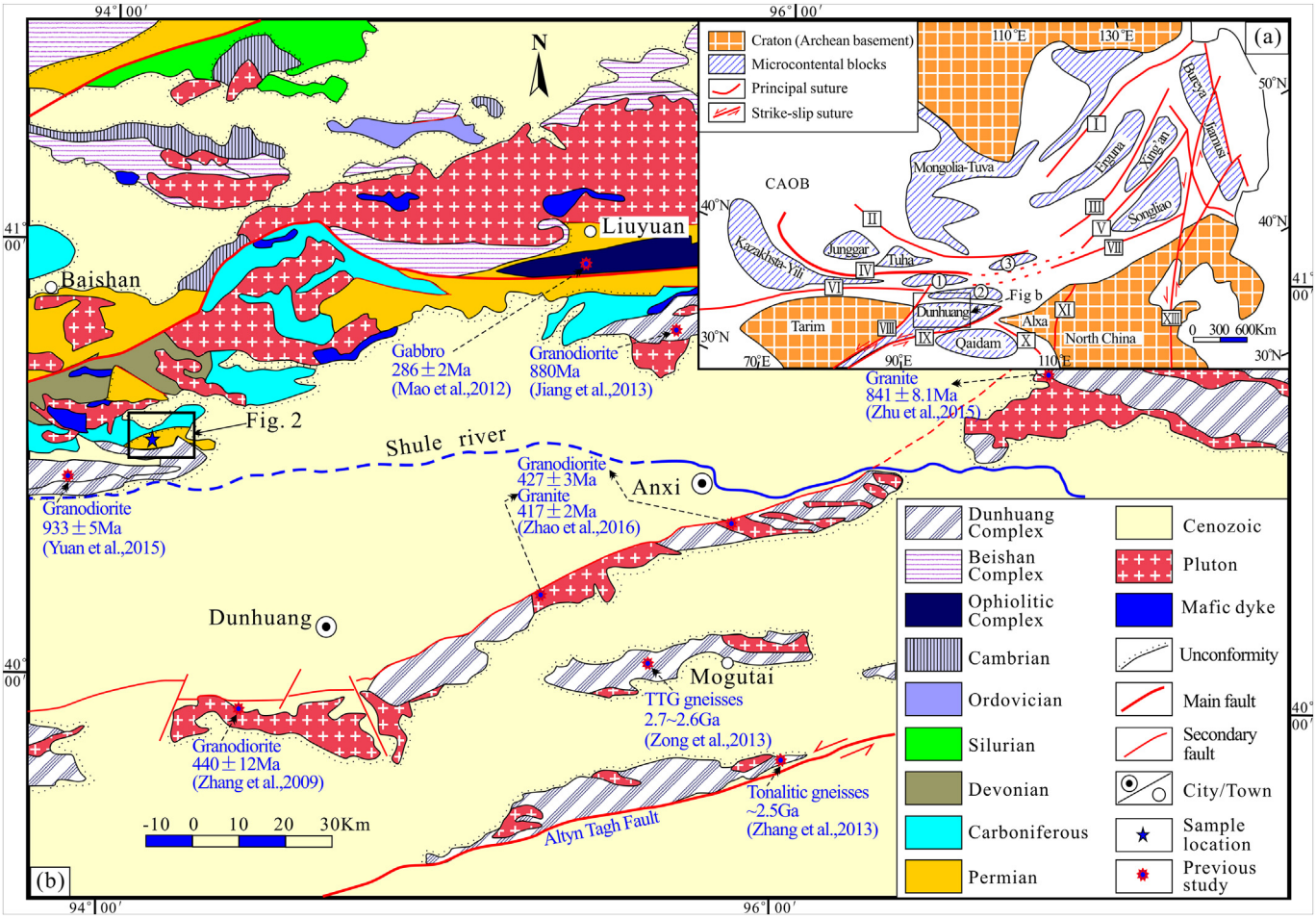


Fig. 1. (a) Tectonic sketch map of Central Asia and location of the study area (modified from Zhou et al., 2018). Microcontinental blocks: ① Central Tianshan block; ② Beishan block; ③ Chaganwule block. Suture zones and Faults: I – Mongol–Okhotsk; II – Erqis–South Mongolian; III – Toudaoqiao–Xinlin; IV – North Tianshan; V – Hegenshan–Heihe; VI – South Tianshan; VII – Solonker–Xar Moron–Changchun; VII – Qiemo-Xingxingxia; IX – Altyn Tagh; X – Longshoushan; XI – Bayanwulashan-Langshan. (b) Regional geological map of the northern part of the Dunhuang block (DHB) and adjacent areas (modified from H.L. Wang et al., 2007). The ages marked inside the map are zircon U-Pb ages from previous studies (Zhang et al., 2009, 2013; Mao et al., 2012; Jiang et al., 2013; Zong et al., 2013; Yuan et al., 2015; Zhu et al., 2015; Zhao et al., 2016). The blue star shows the site of sampling and the box shows the location of Fig. 2(a).

intermediate micro-continents (Şengör et al., 1993; Mossakovsky et al., 1993; Windley et al., 2007; Xiao et al., 2010a). The occurrence of many world-class mineral deposits in the CAOB has made it known as one of the world's most famous large metallogenic domains (LMD) (other LMDs include the Circum-Pacific metallogenic domain and the Tethyan metallogenic domain). For example, it has been described as the "Asian gold belt" in some literature (Xue et al., 2014; C.J. Xue et al., 2016). The CAOB provides not only abundant invaluable resources, it is also an excellent "natural laboratory" to study earth's crust and mantle evolution, continental accretion, collisional process and relationship with metallogenesis. The study of CAOB orogenic and metallogenic processes hence has become one of the most active frontiers in modern international geoscience (Yakubchuk et al., 2001; Windley et al., 2007; Xiao et al., 2003, 2010a, 2010b; Li et al., 2009, 2013). It has been generally accepted that the orogenic belt consists of various tectonic sub-units (e.g., multiple micro continental blocks, island arcs, oceanic islands, accretionary wedges, and so on), with a long history of tectonic evolution (more than one billion years) involving complex subduction and accretion (Khain et al., 2002; Xu et al., 2014; Xiao et al., 2010a, 2010b; Li et al., 2006, 2009). However, there are still hot debates on some fundamental issues. Among the unresolved questions, one notable point of debate is concerned with the timing of the final closure of the Paleo-Asian ocean. Currently, there are at least three different scenarios available: (1) before Carboniferous (Şengör et al., 1993; Xia et al., 2004; Xu et al., 2013); (2) end-Permian to early Mesozoic (Xiao et al., 2009a, 2010a, 2015, 2018; Yi et al., 2015; Huang et al., 2018; Zhang et al., 2018a, 2018b); and (3) Late Carboniferous-Middle Permian in the west whereas Late Permian-Middle Triassic in the east (Shi, 2006; Li et al., 2009, 2013; Zhang et al., 2008; Wei et al., 2016).

The Dunhuang block (DHB) is located in the south margin of the CAOB, between the Tarim block and NCB (Fig. 1a). As such, it occupies a geological area that is critical to understand the tectonic evolution of the Paleo-Asian ocean in connection with the geological history of the Tarim block and NCB. To date, the tectonic affinity and identity of the DHB remains a controversial issue: it is considered by some as part of the Tarim block on the one hand (Mei et al., 1998; Zhang et al., 2011; Meng et al., 2011), while some other geologists have argued that they represent two separate blocks because they are divided by the Qiemo-Xingxingxia fault (QXF) (Zhu, 1989). A third scenario has also been proposed: based on isotopic ages of basement metamorphic rocks and geophysical information, some workers suggest that the DHB pertains to the NCB rather than the Tarim block (Zhang et al., 1997; Zhao et al., 2015). Moreover, the recent discovery of significant Silurian-Devonian metamorphic rocks from the DHB (Zong et al., 2012; Wang et al., 2017, 2018) and the absence of similar rocks in the Tarim block or NCB raises one further possibility, that the DHB block might also have existed and behaved as an independent micro block or amalgamated tectonic mélange throughout the Paleozoic. Notwithstanding these

diverse views, interestingly, as far as the Permian is concerned, none of these models have been tested with paleomagnetic data despite its undisputed role for tectono-paleogeographic reconstructions. Understandably, this is because until now, no Late Paleozoic paleomagnetic data have been published to constrain the paleogeographic position of the DHB at these times.

To our knowledge, this paper is the first study of Early Permian paleomagnetism from the DHB; it provides fresh and coupled geochronologic and paleomagnetic data that are crucial for constraining the timing of the closure of the Paleo-Asian ocean in the Beishan area, as well as information on the relative position of the DHB in relation to several adjacent tectonic blocks.

2. Geologic setting, sampling and methods

The DHB is triangular in shape (Fig. 1a), separated from the Tarim block by the Qiemo-Xingxingxia fault (QXF) to the northwest, from the Qaidam block by the Altyn Tagh fault to the southeast, and connected with the Beishan orogenic belt by the Liuyuan ophiolite to the north (Xiao et al., 2010b; Yang et al., 2012; Li et al., 2009, 2013). The

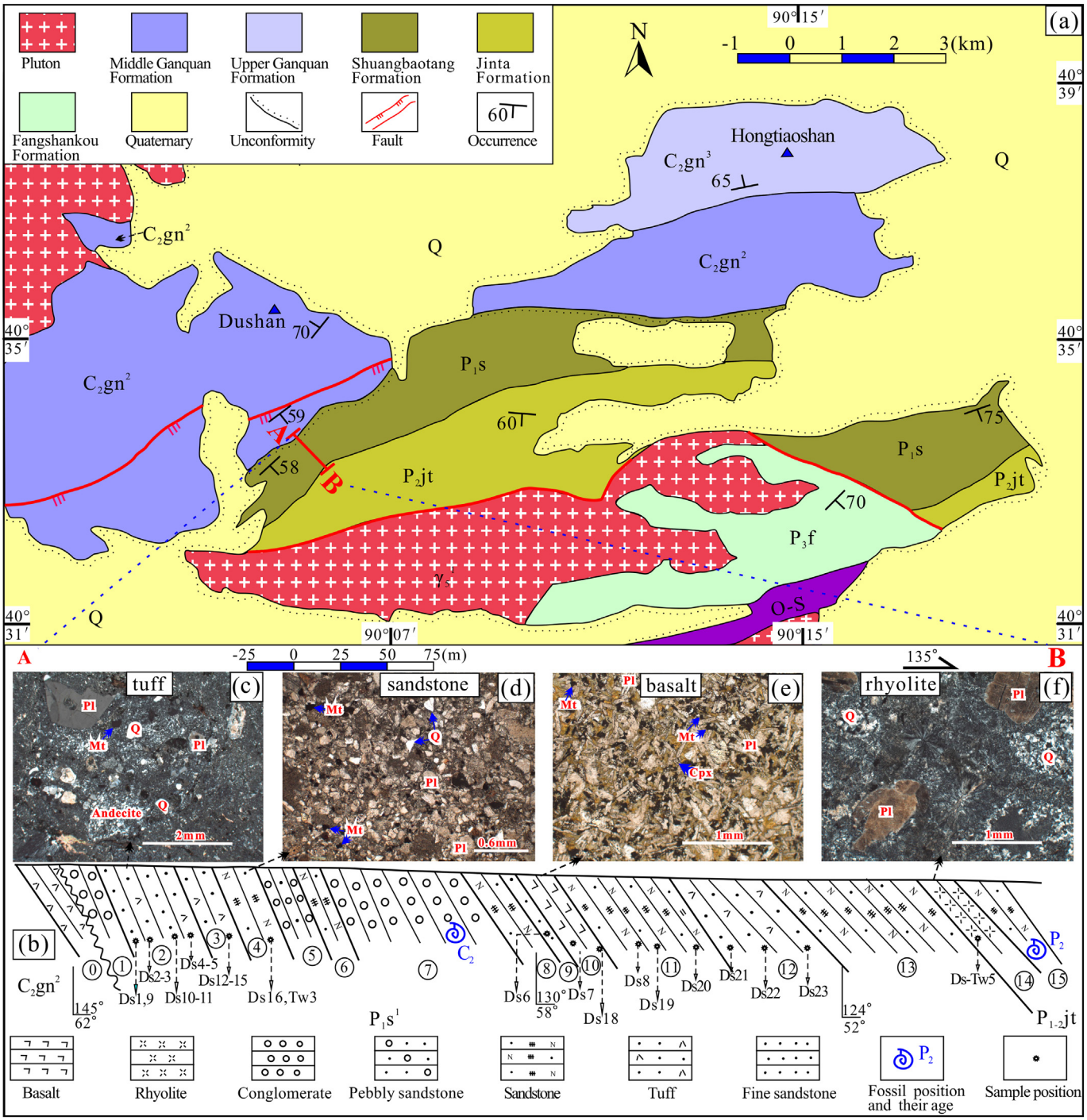


Fig. 2. (a) Simplified geological map of the investigated area (modified after BGMRGS, 1974); (b) Measured geological section of the Shuangbaotang Formation, showing details of sampling locations, site distributions, fossil positions, dip directions and dip angles for each sampled site; (c–f) Microscopic observation results for studied samples. Mt: magnetite, Q: quartz, Pl: plagioclase, Cpx: pyroxene.

oldest rock outcrop of the DHB (Dunhuang Complex) is dated from 840 Ma to 2.7 Ga in this region (Fig.1b; Zhang et al., 2013; Zong et al., 2013; Yuan et al., 2015; Zhu et al., 2015; Jiang et al., 2013), suggesting the DHB has undergone a long geological evolution history.

The sampled locality lies at the north-west margin of the DHB, 50 km northwest of Dunhuang city (40.56°N/94.08°E) (Fig. 2a). A simplified stratigraphic profile of the measured sections is shown in Fig. 2b. The samples came from the Shuangbaotang Formation which unconformably overlies Upper Carboniferous (Ganquan Formation) tuff and is conformably overlain by Mid-Permian (Jinta Formation) rhyolite. The Shuangbaotang Formation consists of tuff (Fig. 2c), sandstones (Fig. 2d), basalt flows (Fig. 2e) and conglomerates, with a variable total thickness ranging from 2 m to 80 m in the study area; it contains two fining-upward lithologic cycles. The age of this formation is generally regarded as Early Permian, based on its own marine fossils (Shi et al., 1995; Shen et al., 2006; Niu et al., 2018), as well as constraints from the Late Carboniferous fossils found from the pebbles embedded in the conglomerate beds in the lower part of the formation, and Mid-Permian fossils from the overlying Jinta Formation (BGMRGS, 1974).

Altogether two basalt sites, 11 tuff sites, and nine sandstone sites were sampled from the measured section (Fig. 2b, Table 1). Ten samples were generally collected at each site by a gasoline-powered drill, and oriented by magnetic compass measurements. In addition, two fresh samples were respectively collected for geochronologic dating to further constrain and refine the age of the Shuangbaotang Formation: one from the lower sandstone bed (Ds-Tw3) (Fig. 2d) and the other from rhyolites (Ds-Tw5) (Fig. 2f) in the upper of the studied section.

All paleomagnetic samples were measured at the paleomagnetic laboratory of the Institute of Geomechanics, Chinese Academy of Geological Sciences in Beijing. Therein, one or two 2.2 cm long cylindrical specimens were sliced from each sample for further rock magnetic and paleomagnetic measurements. All samples were treated with stepwise (15–18 steps) thermal demagnetization in an ASC TD-48 oven with an internal residual field <10 nT. A 2G cryogenic magnetometer was used to measure the remanent magnetization, the directions of the remanent component were separated by the principal component analysis (Kirschvink, 1980), and the site-mean remanence directions were calculated using Fisher statistics (Fisher, 1953). The anisotropy of magnetic susceptibility (AMS) of samples was measured by a KLY-3S Kappabridge susceptibility system. In order to characterize the remanence carriers, the acquisition experiments of isothermal remanent magnetization (IRM) were conducted using an ASC scientific impulse magnetizer model IM-10-30. Thermal demagnetization of a three-component IRM with fields of 0.12 T, 0.4 T and 1.2 T was implemented (Lowrie, 1990). Also, thermomagnetic analysis was performed for selected samples by a Curie balance in air (using a KLY3 Kappabridge susceptibility meter coupled with a CS3 furnace).

Zircon grains used for geochronology were extracted by heavy-liquid and magnetic techniques, further purified by hand-picking under a binocular microscope. They were set in an epoxy mount and polished down to half section. The zircon LA-ICPMS U-Pb isotope analyses were performed at the Xi'an Center of Geological Survey, China Geological Survey. Cathodoluminescence (CL) images were taken via the scanning electron microscope in order to identify the internal structures

Table 1
Site-mean paleomagnetic directions of the higher temperature component (HTC) from the Early Permian rocks in the north west of the Dunhuang block (40.55°N, 94.08°E).

Site	S_{lat} (N)	S_{lon} (E)	S/d (°)	−/+	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)	P_{lat} (°)	P_{lon} (°)
Sedimentary rock (sandstone and sedimentary tuff)												
DS1	40°33′45.01″	94°04′51.03″	45/59	8/0	165.8	5.0	179.9	−42.9	75	6.4	−74.4	94.4
DS2	40°33′44.77″	94°04′51.58″	43/55	8/0	164.0	−0.4	179.6	−44.9	141	4.7	−75.9	95.6
DS3	40°33′44.62″	94°04′51.69″	43/55	6/0	166.4	−0.2	184.0	−44.3	90	7.1	−75.0	80.0
DS4	40°33′44.57″	94°04′52.29″	35/59	5/0	162.1	−0.1	180.9	−43.2	222	5.2	−74.5	91.0
DS5	40°33′44.24″	94°04′52.57″	39/59	4/0	164.7	−0.6	183.9	−44.6	116	8.6	−75.3	80.2
DS6	40°33′41.79″	94°04′55.92″	25/51	4/2	342.4	6.2	5.2	43.5	28	12.8	74.2	257.0
DS8	40°33′37.47″	94°05′03.33″	34/51	8/0	158.0	0.3	170.5	−42.0	95	5.7	−71.8	123.0
DS9	40°33′37.48″	94°05′03.34″	46/58	8/0	164.8	1.5	180.7	−46.8	185	4.1	−77.4	91.3
DS10	40°33′45.01″	94°04′51.06″	46/58	6/0	164.9	1.1	181.2	−47.1	231	4.4	−77.6	89.2
DS11	40°33′45.04″	94°04′51.55″	38/62	8/0	164.3	1.1	184.2	−44.6	397	2.8	−75.2	79.2
DS12	40°33′44.89″	94°04′51.88″	38/62	8/0	163.1	−3.7	176.6	−44.9	291	3.3	−75.6	106.0
DS13	40°33′44.73″	94°04′51.60″	45/49	10/0	165.2	−0.8	179.9	−42.7	247	3.1	−74.2	94.4
DS14	40°33′44.30″	94°04′51.71″	38/61	11/0	162.1	−0.3	182.7	−46.6	355	2.4	−77.1	83.4
DS15	40°33′44.71″	94°04′52.40″	38/61	8/0	162.2	0.9	181.6	−45.8	434	2.7	−76.6	87.9
DS16	40°33′44.37″	94°04′52.59″	35/57	12/0	163.5	0.2	180.4	−40.8	365	2.3	−72.7	92.9
DS19	40°33′37.75″	94°05′04.68″	37/51	6/0	162.4	−3.7	179.5	−43.4	113	6.3	−74.7	95.8
DS20	40°33′35.94″	94°05′04.54″	37/52	3/0	171.0	−6.1	189.6	−38.3	162	9.7	−69.3	68.5
DS21	40°33′35.92″	94°05′05.36″	34/52	11/0	168.3	−8.1	188.7	−40.2	426	2.2	−70.8	69.0
DS22	40°33′35.83″	94°05′05.64″	34/52	7/0	164.9	−4.1	182.0	−39.7	410	3.0	−69.5	86.3
DS23	40°33′34.95″	94°05′06.08″	34/52	6/0	162.8	−9.1	182.6	−43.8	84	7.4	−74.9	85.1
Mean				20	344.1	1.7			330	1.8		
							1.7	43.6	449	1.5		
Corresponding pole: 74.8°N/268.2°E; dp/dm = 1.2°/1.9°												
Basalt												
DS7	40°33′39.56″	94°05′00.37″	40/58	5/1	345.6	−6.5	0.2	40.6	56	9.1	72.6	274.0
DS18	40°33′39.51″	94°05′00.34″	37/52	6/1	349.3	0.8	1.9	35.9	20	13.8	69.2	269.0
Mean				2	347.5	−2.9						
							1.1	38.3				
Corresponding pole: 70.9°N/271°; dp/dm = 7.5°/12.7°												
Mean of all sites (Sedimentary rock + Bsalt)	22	344.4	1.3			294	1.8					
							1.7	43.1	403	1.5		
Mean of VGPs		22			K = 386	A_{95} = 1.6	74.5	268.5				

S_{lat} and S_{lon}: latitude and longitude of the sampling site; S/d: dip direction of beds-90°/the mean dip and dip azimuth of site beds; −/+, number of ChRM directions that display reversed/normal polarity in calculating Fisher site-means; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigraphic coordinates respectively; $\kappa(\alpha_{95})$: the best estimate of the precision parameter (radius lies within 95% confidence) for the mean of direction in stratigraphic coordinates; $K(A_{95})$: the best estimate of the precision parameter (radius lies within 95% confidence) for the VGP; dp/dm: semi-axes of elliptical error of the pole at a probability of 95%; P_{lat} and P_{lon}: latitude and longitude of pole in stratigraphic coordinates. (1). Fold tests for all the sites are positive according to McFadden method (1990) (Xi1 IS = 7.729, Xi1 TC = 0.9176, critic Xi at 95% = 7.662). (2). Reversal test for all the sites is positive: Statistics R = 0.11, average Gamma = 3.6°, F (36, 4) = 2.63, Critical Gamma = 4.4°, Critical R at 95% = 0.16, classification A (McFadden and McElhinny, 1990); bootstrap reversal test is also positive (Tauxe et al., 2016).

and to ensure a good selection of analytical sites. The analyses were obtained using a laser-ablation microprobe (GeoLas Pro) coupled to an Agilent (7700×) ICP-MS. Zircon U-Th-Pb measurements were operated on 23 μm diameter regions of single grains. Detailed analytical procedures and experimental parameters follow the instruction given by Yuan et al. (2004). NIST 610 glass was used as an internal standard for U, Th and Pb analyses and zircon 91500 was used as the external calibration standard, which was analyzed twice every six sample analyses. The measured Pb isotopic compositions were corrected for common Pb by means of the measured ^{208}Pb following Andersen (2002). Isotopic ratios and element concentrations were calculated using ICPMSDataCal software (Liu et al., 2008, 2010). Concordia diagrams were obtained using the ISOPLOT software (Ludwig, 2003).

3. Experiments and results

3.1. U-Pb dating

The analytical results of the sandstone sample (Ds-Tw3) and rhyolite sample (Ds-Tw5) are shown in the Supplementary Table 1. Seventy-two grains were analyzed from Ds-Tw3 with reporting uncertainties of individual analyses at the 1σ level, of which 62 were ≤10% discordant (Fig. 3a). The age distribution shows prominent peaks at 432 Ma, 930 Ma and 2540 Ma (Fig. 3b), whereas 12 concordant ages yield a weighted mean ^{206}Pb/^{238}U date of 291.4 ± 2.6 Ma (Fig. 3a). Therefore, the maximum age of deposition for the Shuangbaotang Formation is the Early Permian, strongly corroborating the biostratigraphic age given by the fossils. The zircons indicating the older age peaks (2540 Ma, 930 Ma and 432 Ma) are mostly round in shape (Fig. 3c), suggesting that they had been subjected to weathering and transportation (reworking). These ages are also consistent with previously published geochronological data for DHB basement rocks (Fig. 1b; Zhao et al., 2016). It is therefore very reasonable to consider that these ages represent the age of the DHB basement rocks, which supplied the sediment deposited in the Early Permian. In other words, the Shuangbaotang Formation was deposited when the study area was already part of the DHB during the Early Permian.

Thirty-six grains were analyzed from Ds-Tw5 with reporting uncertainties of individual analyses at the 1σ level (Fig. 3d), of which 29 were ≤10% discordant. Most grains demonstrate features of idiomorphic crystals with oscillatory zoning (Fig. 3f) and present high Th/U ratios (between 0.50 and 1.20), which are characteristic of magmatic zircons (Belousova et al., 2002; Wu and Zheng, 2004). Twenty-eight concordant ages yield a weighted mean ^{206}Pb/^{238}U age of 280.6 ± 2.9 Ma, MSWD = 0.39, which is interpreted as the crystallization age of the rhyolite (Fig. 3e). Only one analysis of zircon grains with inherited core and reaction rim yielded an older ^{206}Pb/^{238}U ages (301 ± 11 Ma), here interpreted as xenocrysts. Thus, taking into account the ages from both Ds-Tw3 and Ds-Tw5, the age range of the Shuangbaotang Formation is considered to be between 291.4 ± 2.6 Ma (lower part) and 280.6 ± 2.9 Ma (upper part).

3.2. Laboratory paleomagnetic analyses

Laboratory measurements of rock magnetism were undertaken to identify magnetic minerals within representative samples. Temperature dependences of magnetic susceptibility (k) experiments obtained from all samples show a sharp drop in magnetic susceptibility at about 580 °C in the heating curve (Fig. 4a–d), indicating magnetite being the main magnetic mineral. In addition, sandstone sample DS9-8 in Fig. 4b gives some evidence for a Hopkinson peak near 580 °C and shows a significant drop between 320 °C and 340 °C, indicative of single domain

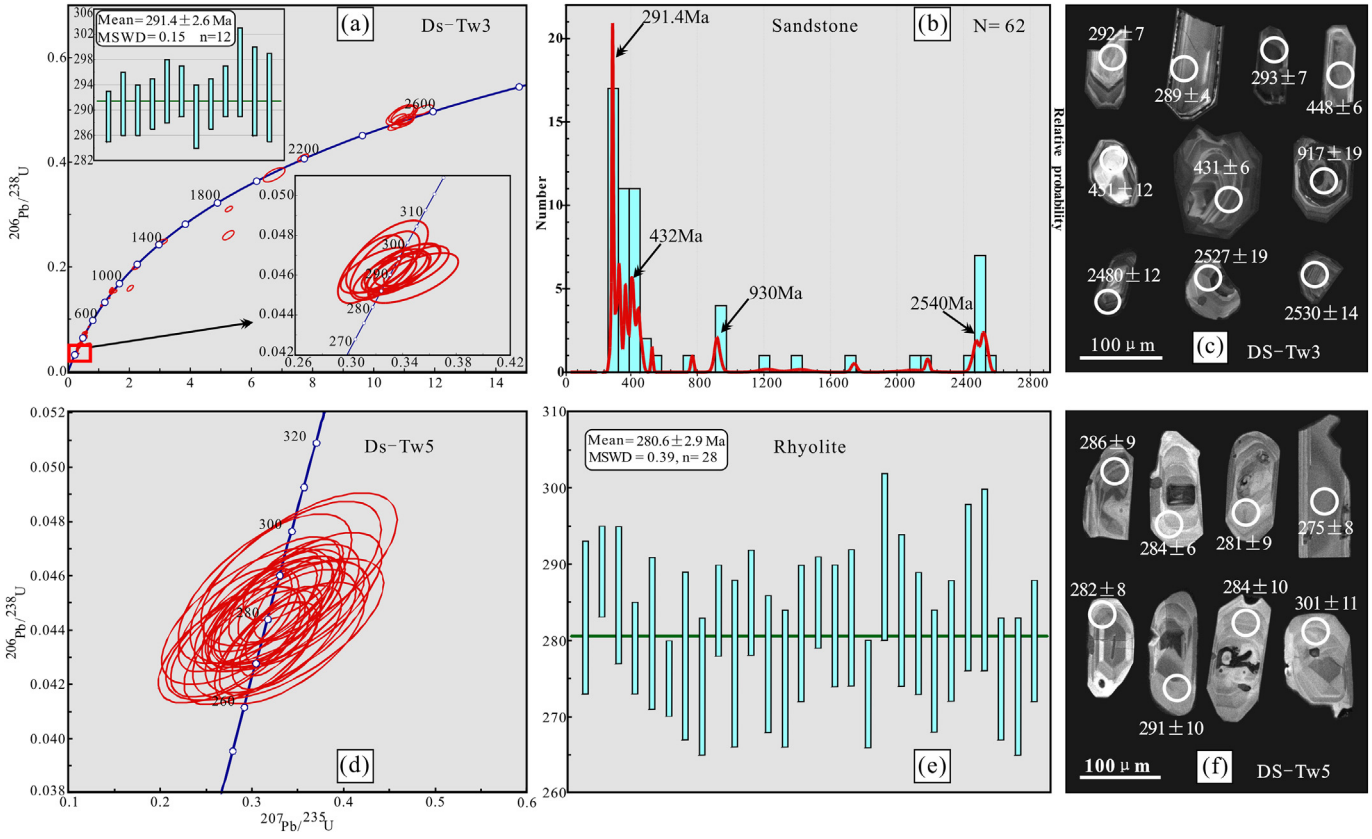


Fig. 3. (a) U-Pb concordia diagram for all analyzed zircons and weighted mean age for the youngest ages and (b) U-Pb cumulative age probability plots of detrital zircons from Sample Ds-Tw3; (c) and (f) are Cathodoluminescence (CL) images and LA-ICP-MS U-Pb (small circles) spots with ages of representative zircons from Sample Ds-Tw3 and Ds-Tw5 respectively; (d) U-Pb concordia diagram for all analyzed zircons and (e) Weighted mean age for the most ages for Sample Ds-Tw5.

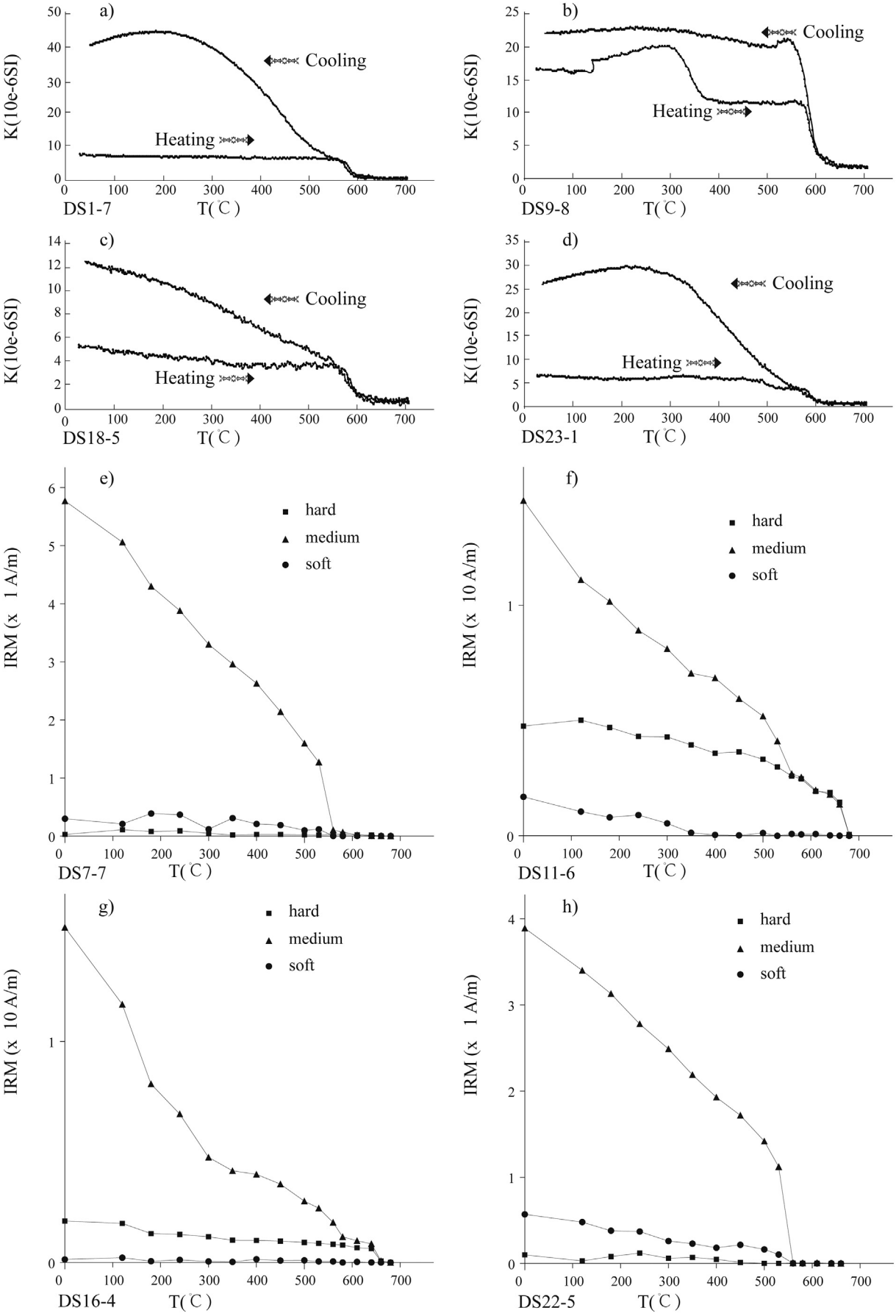


Fig. 4. Thermomagnetic curves of susceptibility (a-d) and thermal demagnetization of three-component IRM (e-h) for representative sample; (a, b, f, g) typical examples from sandstones; (c, e) typical examples from basalts; (d, h) typical examples from tuff.

magnetite and pyrrhotite respectively (Dunlop and Ozdemir, 1997). All sample magnetic susceptibility values of the cooling curve are higher than the values for the heating curve, suggesting that new magnetite phases were formed during heating in air.

The results of thermal demagnetization of a three-component IRM from basalts and tuff samples indicate that the hard component is the lowest, and the medium and soft components are unblocked until about 580 °C (Fig. 4e, h), also indicative of magnetite. As for the sandstone

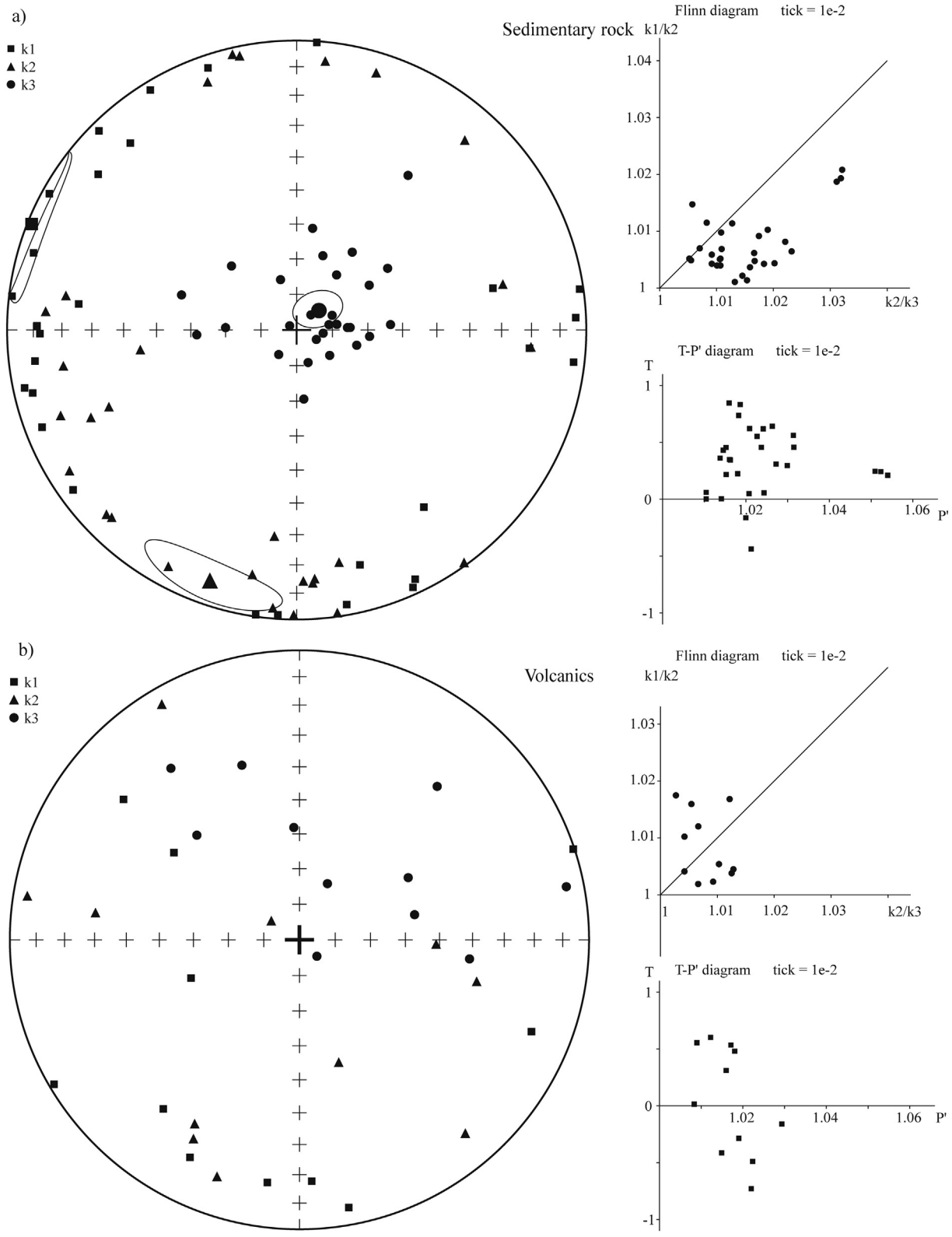


Fig. 5. Anisotropy of magnetic susceptibility data in stratigraphic coordinates from the Shuangbaotang Formation. Lineation = K1/K2, Foliation = K2/K3. (a) Sedimentary rocks (sandstones and tuff rocks); (b) Basalts.

samples (Fig. 4f, g), thermal demagnetization of the soft fraction reveals lower values as well as lower unblocking temperature than the hard and medium components, thus indicative of small amounts of soft magnetic minerals. In comparison, the medium fraction exhibits a significant drop between 530 °C–580 °C, suggesting magnetite, and the hard and medium fractions present a distinct unblocking temperature of 680 °C, indicating the presence of hematite in the sandstone samples.

Regarding the sedimentary samples (sandstones and tuff), the AMS results reveal that the maximum susceptibility axes (K1) are distributed arbitrarily in horizontal surface, and the minimum susceptibility axes (K3) are perpendicular to the bedding plane (Fig. 5a). The corrected anisotropy degree (p′) of the samples mainly ranges from 1.008 to 1.031, and their mean anisotropy degree is 1.016. The sedimentary rocks anisotropy shape factor (T), ranging mainly from 0 to 1, and its Flinn diagram indicate the development of foliation with a dominance of oblateness, thus implying the primary fabric of depositional nature (Tarling and Hrouda, 1993). However, the principal anisotropy directions are dispersed for the basalts (Fig. 5b). In view of such reasonably low degrees of anisotropy, we suggest that neither the basalts nor the sedimentary rocks are likely to have experienced significant strain caused by compaction or tectonic stresses. Meanwhile, microscopic observation of polished thin sections of representative samples indicates the presence of an essentially original magnetic texture, with no chemical alteration caused by fluid activity, or any directionality of the mineral arrangement due to tectonic stresses (Fig. 2c–e). Therefore, the magnetic grains of our samples are considered to have originated from the time of the rock forming, and the Characteristic Remanent Magnetization (ChRM) carried by them thus likely represents their primary magnetization.

Thermal demagnetization was conducted over 15 progressive steps, with representative Zijderveld plots illustrated in Fig.6. The operation of demagnetization on the sedimentary rocks is similar to that of the basalt samples. Two components were separated from the samples: one lower-temperature component (LTC), which was removed below 240 °C, and the other higher-temperature component (HTC), isolated between 380 °C–580 °C in tuff (Fig. 6a, f, h–k, o, q, r) and basalts (Fig. 6g, p), and between 380 °C–580 °C (or 380 °C–680 °C) in the sandstones (Fig. 6b–e, l–n, s–t). The high unblocking temperature (580 °C or 680 °C) also indicates that magnetite dominates in the tuff, basalts and sandstones, but there is a small amount of hematite in some sandstones.

The site-mean directions are listed in Table 1 (the PMSTAT.DIR text is rendered in Research Data part), as well as in Supplementary Table 2 and also illustrated in Fig. 7. The LTC directions yield a site-mean direction of $D_g = 0.9°$, $I_g = 66.5°$, $k_g = 154$, $\alpha_{95} = 2.5°$, and $D_s = 100.1°$, $I_s = 47.5°$, $k_s = 51$, $\alpha_{95} = 4.4°$ after bedding correction (Fig. 7a, b and Supplementary Table 2), with a negative fold test at 99% significance level (k_g/k_s = 3.03, Xi1 IS = 7.593, Xi2 TC = 10.39, Xi2 IS = 2.16, Xi2 TC = 13.53, critic Xi at 99% = 7.662) (McElhinny, 1964; McFadden, 1990). These values indicate that the LTC was acquired after folding, and possibly is a viscous remanent magnetization of the present-day geomagnetic field.

The HTC directions in geographic coordinates are significantly different from the present-day geomagnetic field direction. The site mean HTC direction for the volcanic rocks is $D_s = 1.1°$, $I_s = 38.3°$, n = 2, after tectonic correction ($D_g = 347.5°$, $I_g = -2.9°$, n = 2, in geographic coordinate), and the corresponding paleopole lies at 70.9°N, 271.0°E with dp/dm = 7.5°/12.7°. The site mean HTC direction for the sedimentary samples is $D_s = 1.7°$, $I_s = 43.6°$, $k_s = 449$, $\alpha_{95} = 1.5°$, n = 20, after tectonic correction ($D_g = 344.1°$, $I_g = 1.7°$, $k_g = 330$, $\alpha_{95} = 1.8°$, n = 20, in geographic coordinate), and the corresponding paleopole lies at 74.8°N, 268.2°E with dp/dm = 1.2°/1.9° (Table 1). The inclination values recorded in the sedimentary rocks are consistent with those of the lava flows (Table 1, Fig. 7d).

The degree of inclination shallowing is directly related to the degree of magnetic anisotropy (Jackson et al., 1991). New research shows that a threshold anisotropy of P_{AMS} ~1.04 exists for compaction-induced inclination shallowing in sedimentary rocks in which the remanence is mainly carried by magnetite (Li et al., 2014). The results of our laboratory measurements on rock magnetism have shown that the main magnetic carriers of sedimentary samples studied are magnetite. Considering the low degrees of the anisotropy within the sedimentary samples analyzed ($anisotropy_{mean} = 1.017$) and their consistent inclination values with those derived from the volcanic rocks, we suggest that there is no significant inclination shallowing caused by depositional compaction in the sedimentary rocks from the studied section.

The unfolding process reveals that the site-mean direction for the HTC after tilt correction gives a maximum grouping of direction at ~60% unfolding (D = 169.9, I = −27.7, $a_{95} = 1.4$) (Watson and Enkin, 1993). However, the strata studied in this work all have consistent dips and only minor variations in attitude, which allow the fold test to be applied with small differences in k values ($k_{60\%}/k_{100\%} = 1.26$) and a large confidence interval (ranging from 46.4% to 85.5% and 36.1% to 100.3% at the 95% and 99% confidence level respectively) (Fig. S1). If it was a synfolding magnetization, most likely it could have occurred during the Capitanian stage (late Middle Permian) because of its close stratigraphic association with a megasequence boundary in the study area (Niu et al., 2018). However, on a global scale, most of the Capitanian stage is characterized by a normal polarity (Hounslow and Balabanov, 2016), contrary to the mostly reversed polarity found in our studied samples. In addition, the 60% unfolding data indicates that the DHB (between the Qaidam and Tarim blocks) could have been located at 14.7° ± 0.8°N during the late Middle Permian, but this is impossible. Because the well-studied Tarim block was located at about 30°N (reference point at 39.9°N, 80.2°E) during this time (Zhao et al., 2014), and the Qaidam block (on the southeastern part of DHB), which could not be farther north than the DHB, was located between 25.1°N ± 5.6°–27.7°N ± 5.6° (reference point at 37.3°N, 96.4°E) during the Carboniferous to Permian (Xu et al., 2011; Cao et al., 2017a). Thus, we consider it unlikely that the 60% unfolding suggest a synfolding magnetization.

On the other hand, most of the HTC represent reverse polarity (Normal = 4, reverse = 158), which is in accordance with the Kiaman Reversed Superchron of Late Carboniferous-Permian age. Sites 6–7 and 18 (in layers ⑨–⑩, Fig. 2) show a normal polarity that can be correlated to the Artinskian normal magnetochron CI2n dated at 281.2 ± 2.3 Ma (Hounslow and Balabanov, 2016). For all the sites studied, the angle between the mean normal direction (D = 2.4°, I = 40.0°, n = 3 sites) and the reverse direction (D = 181.5°, I = −43.6°, n = 19 sites) is 3.6°, producing a positive result ($r = 3.6° < r_{critic} = 4.4°$, classification A) (McFadden and McElhinny, 1990). We also used the cumulative distributions of Cartesian coordinates of bootstrapped means to test for a common mean after inverting the reverse directions to their antipodes (Tauxe, 1998). The confidence intervals for the normal and reverse data sets overlap in all three Cartesian components (Fig. S2). This test result indicates that the two mean directions cannot be distinguished at the 95% confidence level and hence they also pass the bootstrap reversals test. Furthermore, the McFadden (1990) correlation fold tests for all the sites yielded Xi1 IS = 7.729, Xi1 TC = 0.9176, critic Xi at 95% = 7.662, in geographic and stratigraphic coordinates, respectively. The result also appears positive. Thus, we suggest that the remanence is likely primary. The mean directions of the basalt series and sedimentary sites can be averaged all together ($D_s = 1.7°$, I_s = 43.1°, $k_s = 403$, $\alpha_{95} = 1.5°$, N = 22) (Fig. 7c, d and Table 1), generating a mean paleopole of site-mean direction-corresponded VGPs at 74.5°N, 268.5°E with K = 386 and $A_{95} = 1.6°$. Consequently, it can be concluded that the paleomagnetic directions from the Early Permian sedimentary rocks and basalts can be regarded as representing primary magnetization; thus, these ChRM directions are appropriate for tectonic reconstructions.

4. Discussion

4.1. The tectonic affinity of the DHB during the Early Permian

The available Late Paleozoic paleomagnetic studies from the CAOB and surrounding blocks (Siberia, Mongolia-Tuva, Songliao, Junggar,

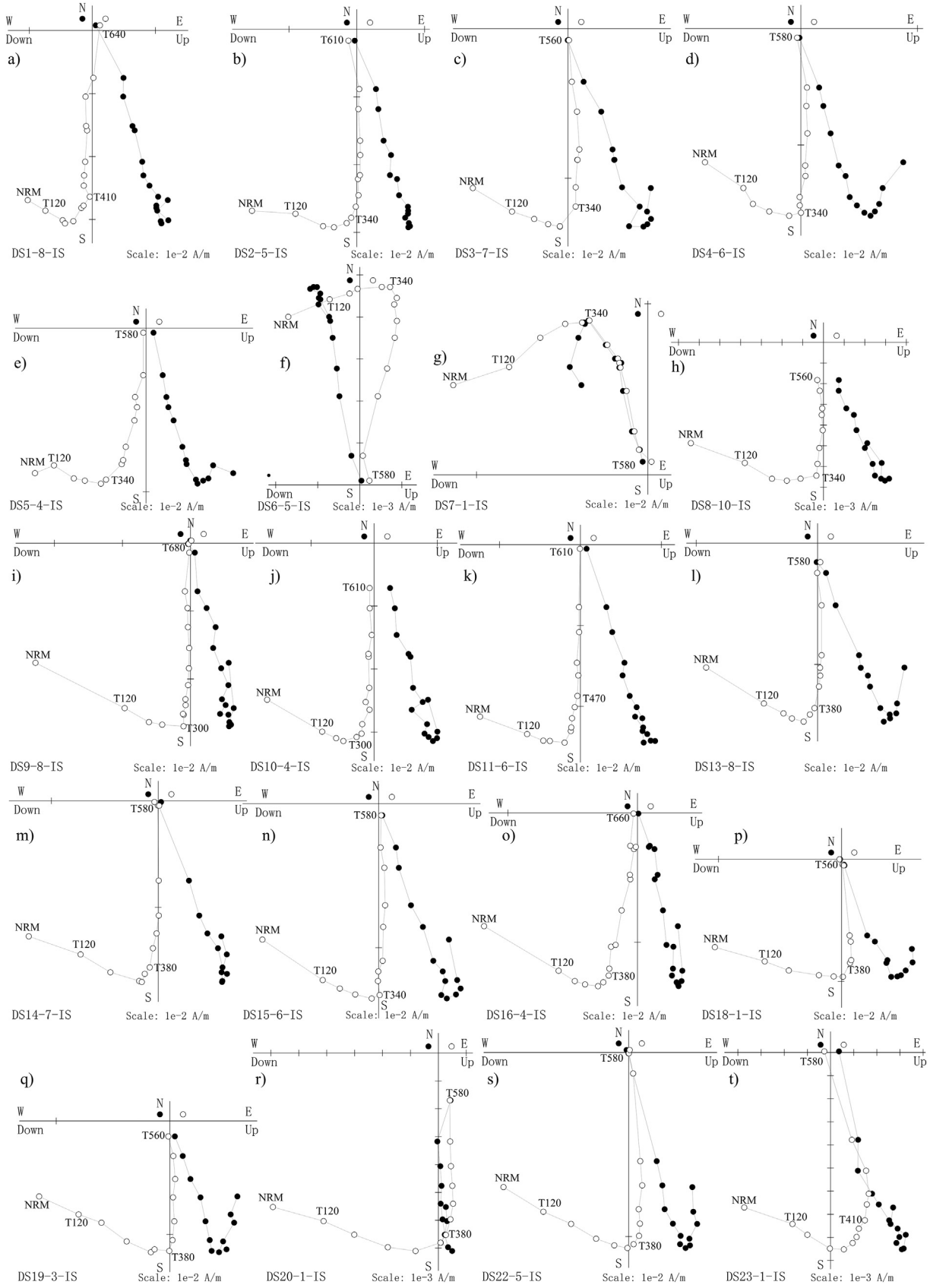


Fig. 6. Orthogonal demagnetization diagrams of representative rocks in situ coordinates. (a, f, h, i, j, k, o, q, r) Typical examples from sandstones; (g, p) typical examples from basalts. Others for tuff rocks. NRM is the natural remnant magnetization.

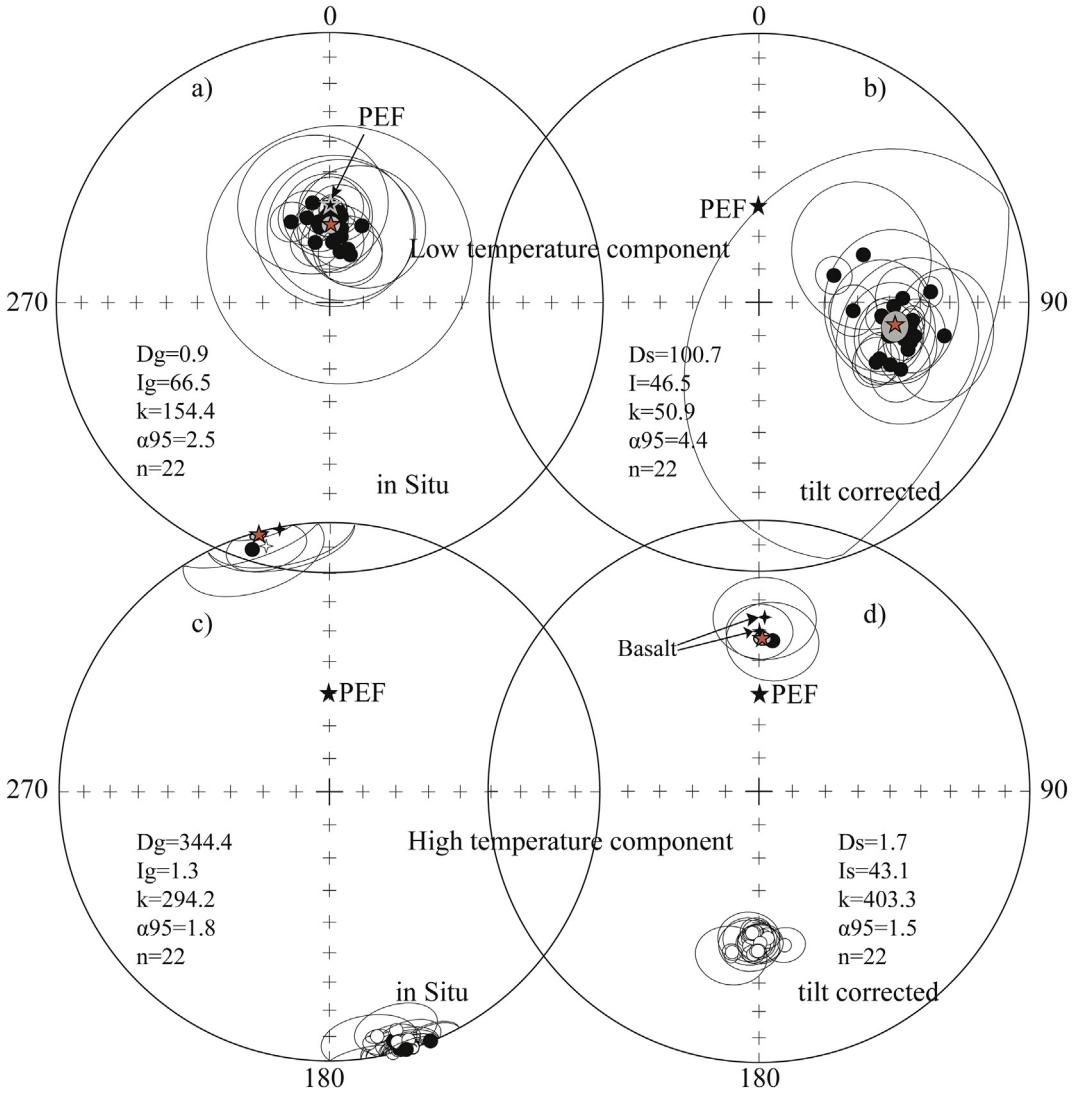


Fig. 7. Equal-area projections of the lower temperature component (LTC) (a–b) and higher temperature component (HTC) for all samples (c–d). The red and black stars represent the mean direction of all sites and the present Earth's field direction (PEF) at Dushan (40.55°N, 94.08°E) (Fig. 2a) respectively.

Kazakhstan-Yili, Tarim, Erguna, Xingan, North China, Qaidam, Dunhuang, Alxa, North China and South China) are listed in Table 2, with corresponding paleopoles shown in Fig. 8. The corresponding paleolatitudes were calculated as a reference point in our studied section (40.55°N, 94.08°E). The small circle passing over the Early Permian pole for the DHB is fit at the same reference point with a paleolatitude of 25.1°N ± 1.2°.

The paleomagnetic studies of the Tarim block began in the 1980's, but only a limited number of reliable Early Permian paleomagnetic data have been obtained with positive fold tests (Bai et al., 1987; Sharps et al., 1989; Gilder et al., 1996; Fang et al., 1998). The averaged Early Permian paleomagnetic pole of these data is at 69.3°N, 226.4°E with α95 = 4.2 (Zhao et al., 2014), which is distinguished from the Early Permian pole for the DHB within a 95% confidence level (this study, Fig. 8). In addition, previous paleomagnetic data indicates that the Tarim block was located at 38.4° ± 5.6°N during the Early Permian (reference point: 40.55°N, 94.08°E) (Table 2). Obviously, within paleomagnetic uncertainties, the DHB shows a significant relative north-south displacement (13.4 ± 5.4°) with respect to the Tarim block (Fig. 8). In other words, we can conclude that the DHB was not part of the Tarim block during the Early Permian, and the relative motion between the two blocks occurred after the Early Permian.

Although the Taklimakan desert (in the Tarim block) covers the wide area of the Qiemo-Xingxingxia fault (QXF) which divides the DHB from the Tarim block, the aeromagnetic data show conspicuously different features between the two sides of the fault (Zhu, 1989). A detailed study of structural deformation has revealed that the QXF, active from late Paleozoic to early Mesozoic, is a giant sinistral strike-slip fault (Li et al., 2009). This revelation corroborates with the paleomagnetic result that indicates that the Tarim has been significantly clockwise-rotated since the Late Carboniferous (Fang et al., 1998). In addition, Early Permian peridotites, gabbros, and marine basalt lavas, all mainly derived from the upper mantle (Su et al., 2013; S.C. Xue et al., 2016), are known to be distributed in the vicinity of the northeastern portion of the fault. Ao et al. (2010) have suggested that the mafic-ultramafic rocks were generated in a subduction-related tectonic setting. This accepted, it would imply that an ocean would have existed between the DHB and the Tarim during the Early Permian.

As can be seen in Fig. 8, there are highly comparable paleo-latitudes between the DHB, Alxa and NCB, despite their significantly different (within 95% confidence intervals) Early Permian poles. The Qaidam block was also located in a similar paleo-latitude zone during the Late Carboniferous to Late Permian (Table 2, Fig. 8) (Xu et al., 2011; Cao et al., 2017a). These comparisons indicate that the DHB might have

Table 2
Available Late Paleozoic paleomagnetic poles for the Dunhuang and adjacent blocks.

Blocks	Age	S_{lat} (°N)	S_{lon} (°E)	N(n)	P_{lat} (°N)	P_{lon} (°E)	A_{95} (dp/dm)	Paleolatitude (°N)	Test	Reference
Siberia (APWP)	C1	–	–	–	22.0	151.0	7.0	38.9 ± 5.5	–	Metelkin et al. (2012)
	C2	–	–	–	35.0	160.0	8.0	38.8 ± 6.3	–	
	P1	–	–	–	42.0	158.0	9.0	43.1 ± 6.6	–	
	P2	–	–	–	46.0	161.0	9.0	42.5 ± 6.7	–	
Mongolia-Tuva	C1(333)	45.0	96.0	22	–	–	–	30.1 ± 4.7	F	Bazhenov et al. (2016)
	C2(315)	43.1	101.0	(42)	32.2	154.6	7.8	41.5 ± 5.9	F	Kovalenko and Chernov (2008)
	P1(275)	45.4	98.8	(53)	71.0	95.0	8.7	59.6 ± 5.1	F	Kovalenko and Chernov (2008)
Songliao	D3	43.5	113.5	3(17)	46.8	349.1	14.6/27.3	19.8 ± 16.5	No	Zhao et al. (2013)
	C2(310)[b]	43.9	120.1	4(29)	32.6	338.1	4.4	4.0 ± 6.2	F	Zhao et al. (1990)
	P1(280)[b]	43.6	117.3	14	41.6	246.1	7.5	−4.0 ± 10.5	F	Zhang et al. (2018b)
	P	43.7	118.4	6(34)	48.7	3.7	5.2/9.1	29.1 ± 8.6	F	Zhao et al. (2013)
	P3-T1(250)	–	–	(62)	64.2	350.6	4.5/7.2	30.6 ± 5.1	F	Didenko et al. (2016)
Junggar	C2	46.5	83.0	12(107)	70.7	209.0	5.0	30.6 ± 4.6	F	Yi et al. (2015)
	P1	43.8	87.8	4(30)	79.5	36.6	6.8	45.6 ± 4.8	F	Choulet et al. (2011)
	P2	43.8	87.8	6(43)	60.4	4.7	5.4	34.7 ± 4.6	F	
Kazakhstan-Yili	C2-P1(305–275)	48.1	75.6	18	42.2	178.8	4.0	29.2 ± 3.8	FR	Abrajevitch et al. (2008)
	P2–3(275–251)	47.9	75.3	19	56.3	180.6	4.5	34.6 ± 5.2	F	
	C2	43.3	83.3	15(98)	68.6	290.6	6.1	19.9 ± 6.9	F	B. Wang et al. (2007)
	P2	43.4	82.5	5(30)	79.7	172	11.3	42 ± 8.4	No	
Tarim	C2	39.9	80.2	8	46.6	170.2	4.4	36.7 ± 3.6	F	Fang et al. (1996)
	P1	–	–	5[a]	59.7	175.1	7.1	38.4 ± 5.6	–	Zhao et al. (2014)
	P2	–	–	5[a]	69.9	188.5	5.4	36.2 ± 4.5	–	
Erguna	C1(363–333)	53.58	127.0	8	39.8	31.6	10.0/15.5	43.3 ± 7.3	F	Kravchinsky et al. (2002)
	C2(320–296)	51.53	115.4	8	10.2	186.2	14.8	5 ± 20.6	F	Xu et al. (1997)
	P2–3	50.65	116.9	14	8.3	183.9	9.5/16.2	5.5 ± 13.2	F	Kravchinsky et al. (2002)
Xingan	C2	45.5	117.2	5	53.4	14.8	9.7	37.3 ± 7.9	R	Chen et al. (1997)
	C2(309–299)	50.1	125.6	11	−17.3	97.1	7.0/12.4	32.1 ± 8.2	No	Zhang et al. (2018b)
Qaidam	C1	37.3	96.4	15(126)	−3.2	147.5	4.8	24.6 ± 5	F	Cao et al. (2017a)
	C2	37.4	96.1	8(61)	−15.1	132.2	5.4	24 ± 5.6	No	
	P3	37.5	99.1	9(56)	64.5	345.4	5.9	28.9 ± 5.6	R	Xu et al. (2011)
Dunhuang	P1(281–291)	40.6	94.1	22(162)	74.5	268.5	1.6	25.1 ± 1.2	F.R	This study
Alxa	C1–2	37.2	105.4	10(39)	10.5	14.0	6.2	14.3 ± 7.7	F	Huang et al. (2001)
	C2	37.0	105.0	7(48)	13.1	11.0	7.0	13.7 ± 8.8	F	Yuan and Yang (2015)
	P1	–	–	(22)	23.9	21.2	12.8	27.9 ± 11.6	F	Wu et al. (1993)
	P2	39.2	106.3	9(68)	27.2	18.8	12.0	28 ± 11.6	F	Yuan and Yang (2015)
North China	C2(310)	–	–	2[a]	31.7	11.1	7.9	24.9 ± 8.1	–	Zhang et al. (2018b)
	P1(290)	–	–	–	37.3	6.5	–	24.8	–	Huang et al. (2018)
	P1(280)	–	–	–	40.3	2.8	–	24.1	–	Huang et al. (2018)
	P1	35.5	110.3	4	47.0	359.2	4.5	25.6 ± 4.6	R	Ma et al. (1992)
	P1–2(270)	–	–	2[a]	48.7	358.0	4.2	25.8 ± 4.2	–	Huang et al. (2018)
	P2	–	–	11[a]	50.3	353.1	5.7	24.1 ± 5.7	–	Yang et al. (1998)
	P2–3(260)	–	–	8[a]	54.3	354.2	4.8	26.9 ± 4.7	–	Huang et al. (2018)
	P3-T1(250)	–	–	11[a]	58.7	358.4	3.7	31.1 ± 3.4	–	Huang et al. (2018)
South China (APWP)	P1(290)	–	–	–	54.8	274.8	–	5.4	–	Huang et al. (2018)
	P1(280)	–	–	–	55.0	271.5	–	5.6	–	Huang et al. (2018)
	P1–2(270)	–	–	6[a]	53.8	248.1	4.3	7.0 ± 5.9	–	Huang et al. (2018)
										(Reference point:40.55N; 94.08E)

Notes: N(n): number of sites (samples); Paleolatitude calculated for the reference point at 40.55 °N, 94.08 °E; F, positive of fold test; R, positive of reverse test.
[a] Denotes the number of paleomagnetic poles used.
[b] Represents results not used in reconstruction, see text for details.

already been amalgamated with the Qaidam, Alxa and NCB before the Early Permian, followed by some significant relative movements between these blocks along some active faults (e.g., the Altyn Tagh fault, Longshoushan fault, Bayanwulashan-Langshan fault, and others) (Chen et al., 2002; Sun et al., 2012; Zhang et al., 2009; Tian et al., 2017).

In summary, we conclude that the DHB was part of the combined Qaidam-Alxa-NCB megablock, rather than part of the Tarim block during the Early Permian. This conclusion is well corroborated by the recent discovery of the Hongliuxia ultrahigh-pressure metamorphism of 428–391 Ma in the boundary zone between the Dunhuang and Qaidam-Alxa-NCB megablock, a tectonic event that has been interpreted to signify the continental collision and suturing between these blocks in the middle Paleozoic (Wang et al., 2017). Accepting these interpretations and also taking into account previous studies concerning the paleo-position of the DHB in earlier Paleozoic times, we propose a sequence of three main tectonic stages concerning the tectonic evolution of the DHB during the Paleozoic: it existed and behaved as a discrete (geographically independent) micro block or a block of amalgamated tectonic mélanges located within the Paleo-Asian ocean during the Early Paleozoic; followed by amalgamation with the combined Qaidam-Alxa-NCB megablock during the middle Paleozoic (amalgamation completed by the Early Devonian); thereafter the DHB and Qaidam-Alxa-NCB megablock formed a single large tectonic block located southeast of, and geographically well separated from, the Tarim block during much of the Late Paleozoic (until at least Early Permian at 280.6 ± 2.9 Ma according to the age obtained in this study).

4.2. Timing of the final closure of the Paleo-Asian ocean

The closure of ancient ocean basins is usually accompanied by crustal shortening due to tectonic convergence. This profound geologic phenomenon is, for example, well demonstrated by the convergence between the Indian and Lhasa blocks over the last 50 million years that has been estimated to be 13.6 ± 6.8° (1510 ± 760 km) (Cao et al., 2017b), accompanied by a further 670–1100 km shortening of the Himalayan crust based on mass-balanced cross-section studies by Decelles et al. (2002) and Guillot et al. (2003). Using a similar approach for estimating crustal shortening in the study area, we estimate that

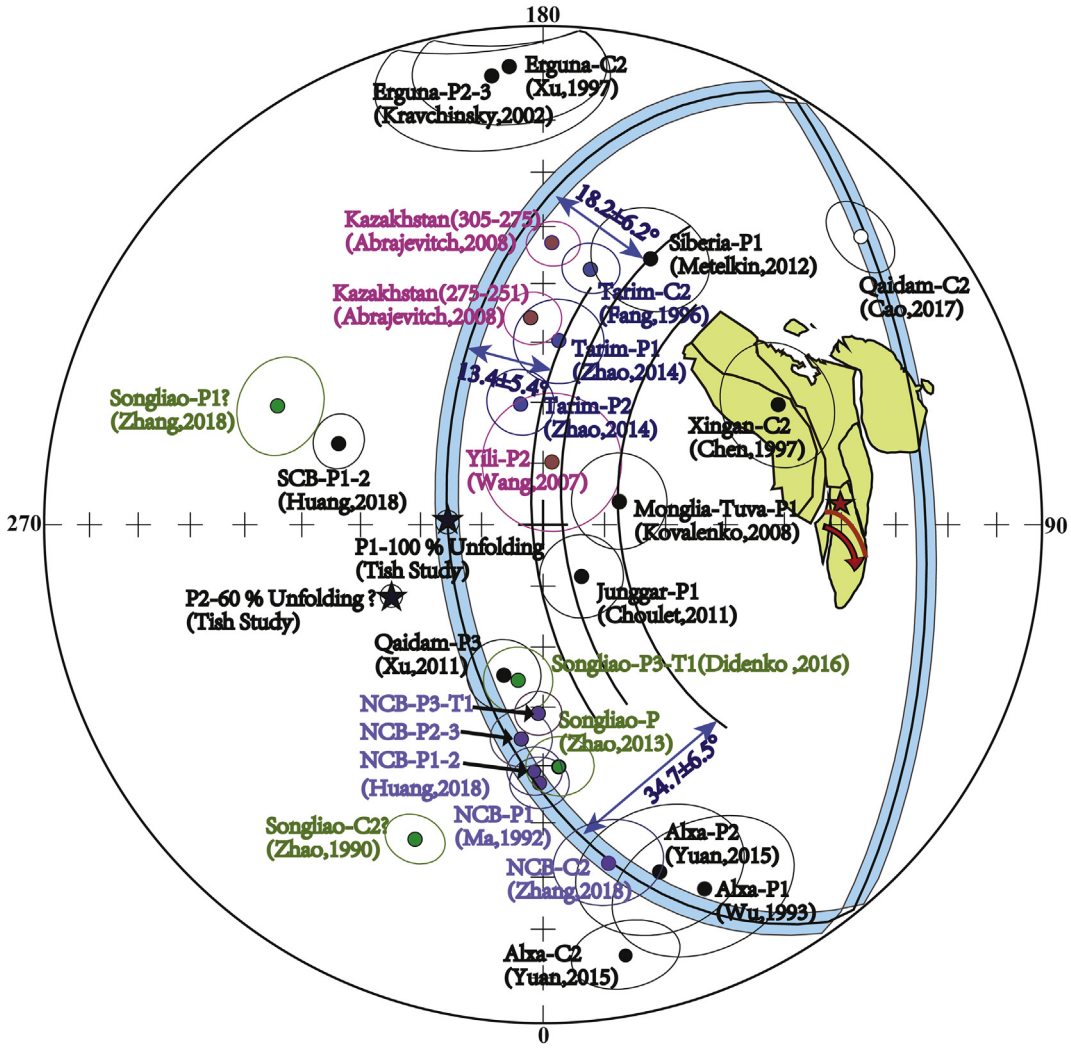


Fig. 8. Comparison of the poles between the DHB and adjacent blocks around the Early Permian. Black star represents the pole for DHB from this study; the red, blue, purple and green dots represent paleomagnetic poles for the Kazakhstan-Yili block, the Tarim block, NCB and Songliao massif respectively; the referenced paleomagnetic poles are listed in Table 2. Question marks (?) represent results not used in reconstruction, see text for explanation.

the DHB might have experienced a latitudinal motion of 20.6° ± 5.2° (2287 ± 577 km), 18.2° ± 6.8° (2020 ± 755 km) and 34.7° ± 6.5° (3852 ± 721 km) with respect to the blocks of Junggar, Siberia and Mongolia-Tuva respectively since the Early Permian (Fig. 8). Such a large amount of N-S convergence exceeds the mere crustal shortening that could be expected as would be in the case of the Himalayan convergence; therefore, an alternative explanation is needed. Here, to account for these large-scale crustal displacements we argue that an ocean perhaps existed between the DHB and Mongolia-Tuva block and extended eastward into the eastern part of the CAOB in NE China during the Early Permian (Fig. 9). The details of this scenario are explained below, and further elaborated in Section 4.3.

For the eastern part of the CAOB, an apparent polar wander (APW) path for NCB has recently been presented by Huang et al. (2018). However, there are only a limited number of Paleozoic paleomagnetic poles for the Songliao-Jiamusi massif, the Erguna and Xingan-Bureya blocks (Table 2), rendering the paleo-positions of these blocks somewhat uncertain for much of the Paleozoic. Independently, Zhao et al. (1990) and Zhang et al. (2018b) investigated the paleomagnetism of the Songliao-Jiamusi massif and concluded that this block was located near the equator during the Late Carboniferous-Early Permian. Paradoxically, however, the Silurian and Carboniferous-Early Permian shallow-marine faunas of this block are consistently characterized by cool-water elements that strongly suggest high-latitude affinities with contemporaneous faunas from Angara, followed by Late Permian floras of mixed Angaran-Cathaysian floristic affinities (Huang and Ding, 1998; Gao and Jiang, 1998; CGU, 2008; Wang et al., 2011; Tian et al., 2016). Numerous previous paleontologic and paleobiogeographic studies (e.g., Tazawa, 1992; Shi et al., 1995; Shi and Zwan, 1996; Shen et al., 2006) have also made similar observations for the Erguna and Xingan-Bureya blocks in regards to their Permian paleobiogeographic affinities. In view of these consistent and strong paleobiogeographic signals, one can convincingly infer that the paleobiogeographic boundary between the Boreal Realm and the Cathaysian Province to lay along the Xar Moron River (which also marks the suture zone between the Songliao-Jiamusi massif and the NCB), at least for the Early Permian. This would imply that there was probably an ocean basin separating the Songliao-Jiamusi massif from the NCB during this time. At the present time, paleomagnetic constraints for the ocean basin is equivocal due to the lack of stratigraphically highly and comparable paleomagnetic data. There is another published and very broadly defined Permian paleolatitude for the Songliao-Jiamusi massif (Zhao et al., 2013). Although it shows some small differences to the multiple, more highly resolved Early, Middle and Late Permian paleolatitudes for the NCB (Ma et al., 1992; Yang et al., 1998; Huang et al., 2018) (Table 2), it is difficult to pinpoint if and when an Early Permian ocean basin existed

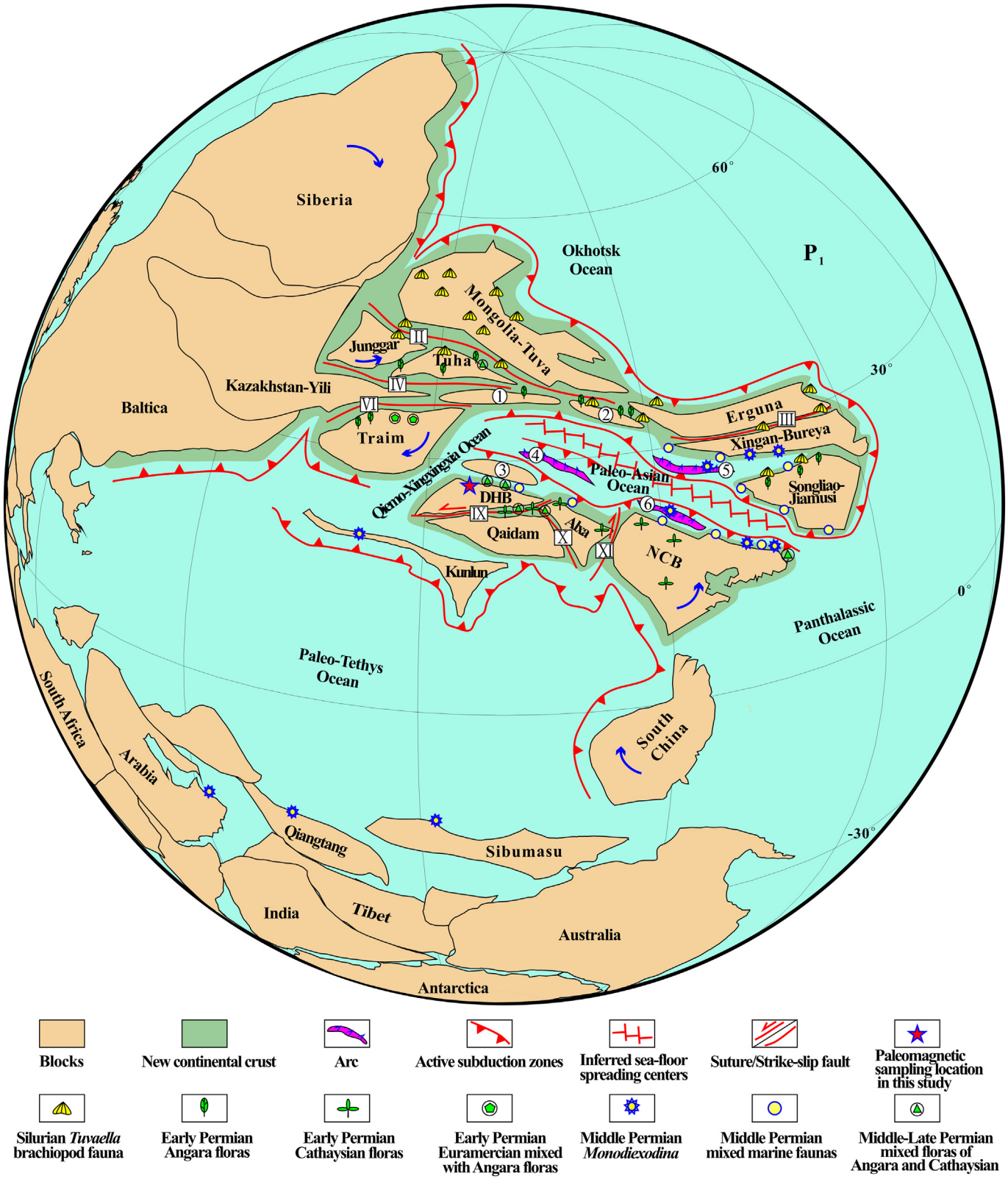


Fig. 9. A schematic paleogeographic reconstruction map of the DHB and adjacent major blocks in Central Asia during the Early Permian (modified from Shi, 2006; Metcalfe, 2013; Yuan and Yang, 2015). The numbers for micro-continental blocks, suture zones and faults are the same as in Fig. 1. Additional blocks are: ④ Dongqiyishan arc, ⑤ Baolidao island arc, ⑥ Bainaimiao arc. Blue arrows show the movement tendency of blocks, with relative longitudes of the blocks unconstrained. The distribution points of Silurian *Tuvaella* brachiopod faunas are based from Rong et al. (1995), Cocks and Torsvik (2007), Wang et al. (2011) and Xiao et al. (2015). The distribution of the Middle Permian mixed marina faunas is modified after Shi et al. (1995). The distribution of Permian floras is based on Huang and Ding (1998), Wang et al. (1999), Zhu (2001), Sun (1995), Sun and Liu (2010), Shen et al. (2006) and Wei et al. (2016). The distribution of *Monodiexodina* is modified after Masayuki and Akihiro (2004), Sun (1985), Sun and Zhang (1988), Sun and Ma (2002) and Ueno (2006).

between the Songliao-Jiamusi massif and the NCB using the paleomagnetic constraints. On the other hand, the Late Permian-Early Triassic poles for the Qaidam block, NCB and Songliao-Jiamusi massif exhibit a high congruency at 95% confidence interval (Fig. 8), suggesting that these blocks had become united during the Late Permian and thereafter have behaved more rigidly as a single block, equivalent to or being integral part of the so-called 'quasi-rigid assemblage' of blocks recently proposed by Huang et al. (2018). In other words, it can be argued that the Paleo-Asian ocean in NE China was closed during the Late Permian, not before or after.

Now, having concluded on the timing of the Paleo-Asian ocean closure in the eastern part of the CAOB, it is necessary to elucidate how this timing extends westward into NW China. In the vicinity of the study area, Permian ophiolites have been discovered in Hongliuyuan (286 Ma, Mao et al., 2012), Enger Us (304 Ma, Zheng et al., 2014); Solonker (288–284 Ma, Jian et al., 2010), Hegenshan (293–298 Ma, Miao et al., 2008), in a west to east direction (present-day sense). These rocks have been commonly regarded as the remnants of paleoceanic crust (Xiao et al., 2009a, 2009b; Li et al., 2013). Likewise, Permian siliceous rocks indicative of deep-sea origin have also been documented in the Beishan-Alxa region (Erduanjing, Hangwula, Enger Us) (Shi et al., 2018; Xie et al., 2014). Within these biochemically deposited siliceous rocks two radiolarian assemblages have been recognized in Enger Us, and dated to be Early Permian and late Middle-early Late Permian respectively (Xie et al., 2014).

Thus, if considered together, the presence of both ophiolites and deep-water radiolaria-bearing siliceous rocks would strongly suggest that an ocean basin existed between the DHB (along with Alxa and NCB) and the Mongolia-Tuva block during the Early Permian (probably into Middle-Late Permian). This inference is also supported by studies from regional structural geology, geochronology and geochemistry, as well as sediment provenance studies (Chen et al., 2000; Xiao et al., 2003; Johnson et al., 2008; Heumann et al., 2012). At the present time, the nature of this Early Permian ocean, especially the timing of its origination and its relationship to the so-called Paleo-Asian ocean, remains unclear. This is because this ocean could be either a newly formed and localized ocean basin, or a remnant of the Paleo-Asian ocean. There is good evidence to support the former view. For example, based on the presence of a major regional unconformity between the Upper Devonian and underlying rocks, and their different deformation patterns, Xia et al. (2007) and Xu et al. (2013) proposed that the Paleo-Asian ocean was closed before the Early Carboniferous, and that the Carboniferous-Permian basic lavas (ophiolites) and radiolarian-bearing siliceous rocks represent products formed in a post-orogenic rift setting (see also Xia et al., 2008; Zhang et al., 2015; Niu et al., 2018). This scenario is further supported by the fact that there are no obvious latitudinal differences between the blocks of Songliao, Alxa, and Erguna since the Late Devonian (Zhao et al., 2013; Huang et al., 2000; Kravchinsky et al., 2002), suggesting that they would have been amalgamated by that time.

In summary, although we infer the existence of an Early Permian ocean between the DHB and Mongolia-Tuva, and between the NCB and Songliao-Jiamusi based on both paleomagnetic and paleobiogeographic evidence, at this stage we cannot rule out the possibility that this ocean might represent a newly opened ocean basin following the closure of the Paleo-Asian ocean during the Late Devonian, and that this newly formed ocean was short-lived only to be closed in the Late Permian (or latest Permian). Obviously, the strength of this hypothesis requires future testing with more high-resolution and high-quality Late Paleozoic paleomagnetic data on the other blocks around the Paleo-Asian ocean.

4.3. Early Permian paleogeographic reconstruction for the Paleo-Asian ocean (sensu lato)

As illustrated in Fig. 8, the Early Permian paleomagnetic pole of the Mongolia-Tuva block is not clearly distinct from the pole for the Junggar block (Choulet et al., 2011) within 95% confidence interval, which indicates that the two blocks had been amalgamated and behaved more rigidly during the Early Permian. The paleomagnetic poles of the Kazakhstan-Yili block (red dots in Fig. 8) are near the poles for the Tarim block (blue dots in Fig. 8) during the Late Carboniferous-Late Permian, suggesting that these blocks had already been amalgamated and behaved more rigidly since the Late Carboniferous. Moreover, the Early Permian paleolatitudes of all these blocks were not distinct from that of Siberia. We therefore suggest that a huge tectonic block likely had been formed by the Early Permian, composed of Junggar, Kazakhstan-Yili, Mongolia-Tuva, Tarim and Siberia. However, the Early Permian poles of Mongolia-Tuva, Tarim, and Siberia are conspicuously different from each other within 95% confidence level due to various mountain ranges or strike-slip fault systems developed within the huge block.

There are no Early Permian paleomagnetic poles yet available for the Erguna and Xingan-Bureya blocks. The Late Carboniferous poles for the Xingan-Bureya block (Chen et al., 1997; Zhang et al., 2018a) are incoherent between these two studies, but both indicate that the block was situated at 28°–30°N during the Late Carboniferous, in agreement with the presence of cold-water Boreal-type faunas in both blocks. On the other hand, evidence suggests that the Erguna block had moved south during the Carboniferous-Permian (Xu et al., 1997; Kravchinsky et al., 2002). Besides, the Silurian *Tuvaella* brachiopod fauna is common in the above two blocks as well as in Mongolia (Rong et al., 1995; Wang et al., 2011; Xiao et al., 2015). All these strongly indicate that the Erguna and Xingan-Bureya blocks formed integral parts of the Mongolia-Tuva block throughout the Early and Middle Paleozoic. Now considering the fact that the Central Tianshan Block (Fig. 1), Chaganwule and the Songliao-Jiamusi massifs all contained a similarly mixed biota (including both floras and marine faunas) by the late Early Permian (Shi et al., 1995; Shen et al., 2006; Tazawa, 1992), these blocks can be considered to have amalgamated into a single integrated tectonic block during the later parts of Early Permian. It is based on this understanding that herein we provide a schematic paleogeographic reconstruction of the 'Paleo-Asian ocean' (sensu *lato*) and surrounds during the Early Permian, as shown in Fig. 9. A notable feature of this reconstruction is the presence of some micro-continents within the Paleo-Asian ocean as well as subduction zones, accretionary prisms and arcs (e.g., Dongqiyishan arc, Baolidao island arc, Bainaimiao arc) (Zheng et al., 2013; Xiao et al., 2015; Ao et al., 2016).

This reconstructed Early Permian paleogeography is in agreement with contemporaneous paleobiogeographic patterns. During the Early Permian, the paleobiogeographic boundary between the Boreal Realm and the Cathaysian Province would have extended along the Beishan-Solonker–Xar Moron–Changchun belt (Fig. 9). In Xinjiang Province (NW China), Early Permian floras from of the Tuha and Tarim blocks contain elements of a mixed Euramerican and Angaran flora but lacking Cathaysian elements (Wu et al., 1997; Wang et al., 1999; Zhu, 2001; Wei et al., 2015, 2016), indicating that Tarim and Junggar were close to, or formed part of, the combined Mongolia-Tuva and Siberia mega-block, but far from the NCB which then incorporated the DHB (Fig. 9). On this basis, we infer a new paleo-ocean, here named the Qiemo-Xinxinxia paleo-ocean, which existed during the Early Permian and separated the DHB and the rest of the integrated Qaidam-Alxa-NCB from the Tarim block (Fig. 9). An independent line of evidence in support of the inferred paleo-ocean comes from the distribution of the Permian fusulinid genus *Monodiexodina*. This genus has only been found in the middle latitudinal zones of both hemispheres during the Early-Middle Permian, thus demonstrating a classic bi-temperate biogeographic distribution pattern (Shi et al., 1995; Shi and Grunt, 2000; Ueno, 2006). This peculiar distribution pattern cannot be properly explained unless a marine biogeographic migration pathway is invoked that had existed between the northern and southern hemispheres. In other words, the inference of the Qiemo-Xingxingxia paleo-ocean would fulfill the requirement of facilitating the bi-temperate distribution of *Monodiexodina*.

As already noted above, the distribution of a biogeographically mixed mid-Permian marine faunas in northern and northeastern China, Russian Far East (South Primorye) and parts of Japan is a very conspicuous feature of the CAOB in the northern part of North China. Detailed studies on the spatio-temporal distribution of these mixed marine faunas by Shi et al. (1995, 2001) have led the conclusion that localities bearing these mixed marine faunas would have been located in a mesothermal zone in the northern hemisphere, estimated to be within the 25°–35°N paleolatitudinal zone. This estimation based on biogeographic features accords well with the calculated Early Permian paleolatitude reported herein for the first time for the DHB (25.1°N ± 1.2°). Notably, in many areas within the CAOB in northern and northeastern China, the mixed mid-Permian marine faunas are succeeded by mixed Angaran and Cathaysian floras, the age of which becomes progressively younger toward the east (Wei et al., 2016). In the study area (eastern Xinjang and Gansu), the oldest occurrence of these mixed floras appeared in the Middle Permian. This would imply that the Qiemo-Xingxingxia ocean between the DHB and Tarim block had already closed by this time. However, further east into eastern and northeastern China, the mixed Angara-Cathaysian floras did not emerge until Late Permian, signaling the final closure of the Palaeo-Asian ocean in this region at this time.

5. Conclusion

We have reported the first paleomagnetic results from 22 sites of Early Permian tuff, basalt flows and sandstones collected from the northwestern part of the Dunhuang block in northwest China. A high temperature characteristic magnetic component with both normal and reverse polarities was isolated by stepwise thermal demagnetization. Positive fold and reverse tests likely suggest a primary magnetization dated between 280.6 ± 2.9 and 291.4 ± 2.6 Ma. Considering the consistent inclination values recorded and the low degrees of anisotropy for all samples, it is interpreted that no significant inclination shallowing has been caused by deposition and/or compaction after the Early Permian in the study area. Based on data from this study as well as previously published Late Paleozoic paleomagetic data from adjacent blocks or areas, it is suggested that the Dunhuang block may have already been amalgamated with North China, and the Alxa and Qaidam blocks during the Early Permian, but was separated from the Tarim block by a small ocean (here named Qiemo-Xingxingxia paleo-ocean) at this time. Using both paleolatitudes and paleobiogeograpohic information, a schematic Early Permian tectono-paleogeographic reconstruction of eastern Asia is proposed. In this reconstruction, the Paleo-Asian ocean (sensu *lato*) is shown to be still open during the Early Permian as a relatively small ocean basin featured with island arcs, subduction zones and microcontinents. In this scenario, the Paleo-Asian ocean is depicted as a remnant ocean basin which was finally closed during the Late Permian. However, it must be pointed out that at the present time, it is not possible to rule out an alternative scenario that the Paleo-Asian ocean was closed before the Late Devonian but reopened in the Early Permian or even earlier.

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Paleomagnetic data were analyzed using K.R. Ludwig (2003), R. Enkin's, J.P. Cogné's (2003) and Tauxe et al.'s (2016) computer program packets.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gr.2018.10.012.

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